

PRODUCT REQUIREMENTS DOCUMENT

Vision360 FSM Platform
Phase 1A · CORE (MVP) · Version 2.0 — July 6, 2026

Project	Vision360 FSM Platform — Phase 1A CORE (MVP)
Based on	BRD v1.2 · Architecture Doc Feb 23 · Product Brief Feb 17 · IA Doc Feb 19 · Wireframes Feb 20 · Discovery sessions · 31 review meetings 2026-04-01 ... 2026-07-06 → DECISIONS.md
Version	v2.0 · July 6, 2026 (<i>supersedes v1.0 · April 2, 2026</i>)
Status	Reconciled against DECISIONS.md — 619 decisions, 55 reversals; meeting #31 (2026-07-06) = newest source, “newest wins”

Change Log — v1.0 → v2.0

Only points that **changed** what v1.0 stated are listed; each cites its decision ID in DECISIONS.md. New reversals found on the full cross-meeting sweep are marked ★.

#	Area	v1.0	v2.0 (final)	Source
1	Platform priority	Mobile-first	Web-first — desktop/web primary (~99% of use); native mobile app = Post-MVP	D-18-10
2	Left navigation	Create + 10 items incl. Calendar, Reports, Settings	9 items: Home · Schedule · Clients · Jobs · Estimates · Invoices · Payments · Expenses · Items	D-09-2/3/4, D-08-15
3	Schedule name / Home	“Calendar (simple)”	Renamed “Schedule” ; “Home” added to nav	D-08-27 → #09
4	Top-bar placeholders	AI / Phone-VoIP / Messaging stubs	Removed from CORE (no “coming soon”)	D-09-3
5	Create / Reports / Settings	in left nav	Top bar: Create (+), Reports (Home top-tab), Settings (gear)	D-09-2/4
6	Back-navigation	—	Breadcrumbs (re-adopted)	D-12-3
7	Client statuses	generic	Active / Prospect / Inactive (+Prospects KPI; auto Prospect→Active on 1st payment)	D-20-1

#	Area	v1.0	v2.0 (final)	Source
8	Client delete	“delete client”	Deactivate only, no hard delete	D-08-16, D-14-19
9	Client type	residential/ commercial	Residential only in create (commercial dropped)	D-08-12
10	Client custom data	—	2 custom fields ; tags = Pro	D-08-7, D-10-16
11	Duplicates	—	Merge (older-ID survives) + Not-a-duplicate; jobs have none	D-20-1/5/8/11
12	Tax config	On Total/Deducted/ Per Item/None	Tax Profiles (dropdown, in Settings)	D-08-23, D-10-3, D-15-2
13	Doc numbering	custom numbering format	System numbering only (custom removed)	#13 (numbering options removed)
14	Roles	solo only	RBAC: Super Admin / Admin / Worker + custom roles , per-module perms	D-19-16
15	Settings layout	flat	Accordion (expandable categories)	D-11-1
16	Appearance / theme	—	Theme (light/dark) removed ; Appearance = sidebar expand/collapse only	D-25-17
17	Estimate options	“up to 4 options”	3 options = 3 tabs, in MVP (good/better/best)	D-31-13 (← D-13-7, D-22-12)
18 ★	Estimate expiration	default same day	Default 30 days, configurable in Settings	D-25-1 (← D-24-3)
19	T&C on estimate	—	Full T&C pages inside PDF + p.1 disclaimer + link in email	D-24-12
20	Signature settings	—	Require before start & on completion ; parent/guardian dropped	D-31-9
21	Item types	—	5: Service · Material · Equipment · Asset · Admin (Fees→Admin); price book = customer-facing view	D-26-8, D-30-12
22 ★	Recurring item charging	—	No per-item “charge per appointment/once/don’t charge” menu — use separate line items	D-29-4 (← part of D-26-17)

#	Area	v1.0	v2.0 (final)	Source
23	Job statuses	—	System (Scheduled/In Progress/Completed) + custom (Canceled, Paused...); unscheduled→scheduled→dispatched→in progress→paused→completed/canceled	D-31-11/12
24	Job pending column	—	“Pending jobs” with Unassigned / Unscheduled chips ; move-off-schedule → unscheduled	D-22-5, D-22-2
25	Schedule board	(was PRO)	Simple board in CORE: ≤3 techs (solo + up to 2 employees), drag-drop, status icons, movable route map, compact rows, no KPI blocks	D-09-5 (3-tech limit), D-27-1/3/6/9
26	Payments	—	Collect (card, processed) vs Record (external + ref #); one payment can settle multiple invoices if each is paid in full , but a partial/deposit = one invoice at a time	D-24-28, D-31-4/5 (refines D-25-7)
27	Invoice statuses	—	Unpaid · Overdue · Paid · Partially paid · Void	D-16-15
28	Expenses	—	Linked to a job (job-costing) or left unlinked (overhead); not to invoices; vendor dropdown; delete on returned purchase	D-25-8, D-25-14, D-29-15
29	Reports	—	Report module (categories, quick filters + custom range, column select, export/generate/print); scheduled email reports in scope ; “Sales report” (was Revenue), “Invoice AR aging” (was Invoice summary)	D-31-6/7/8, D-30-2/4/7
30	Localization	US-only	EN · ES · PT · FR · DE · DA · NL (only Chinese/Arabic out)	D-25-18
31	Native mobile / Project B	mobile app = same product	Native field app = Post-MVP ; separate “Aptora bridge” (Project B) may reuse it	D-13-2

1. Executive Summary

Vision 360 CORE is a web-first Field Service Management (FSM) platform (responsive web in MVP; a native mobile app is a Post-MVP growth feature) purpose-built for solo home service operators — HVAC technicians, plumbers, electricians, roofers, pool cleaners, and similar tradespeople who run their entire business from their vehicle.

1.1 The Problem

Solo operators manage 2-3 jobs daily without administrative support. They juggle customer information, estimates, invoicing, and scheduling using scattered notes, mental math, and fragmented tools. This leads to unprofessional-looking documents, delayed payments, missed follow-ups, and zero visibility into business performance — all while they're trying to focus on the work itself.

1.2 The Solution

Vision 360 CORE puts a complete business command center in the solo operator's pocket. Open the app and see your entire day at a glance. Create professional, branded estimates on-site. Convert accepted estimates into invoices with one tap. Track payments and follow up on outstanding invoices. All orders, all customers, all money — in one place, in the browser.

1.3 Product Vision Statement

"Your day at a glance. Estimates in seconds. Paid before you leave the job site."

1.4 What Makes This Special

Vision 360 CORE is the **first FSM solution designed specifically for solo operators** — a market segment that:

- Enterprise platforms (ServiceTitan, Workiz) ignore due to complexity and pricing
- Simple invoicing apps (Invoice Simple) cannot serve due to lack of job management, scheduling, and workflow integration

This is not a stripped-down version of an enterprise tool. This is a purpose-built application for solo tradespeople and their back-office helper — delivered **web-first** (great on a laptop at the office or kitchen table; a native field app that works from the vehicle follows Post-MVP).

Key "Aha" Moments:

1. *"I opened the app and instantly saw my whole day laid out"*
2. *"I created a professional estimate on-site — all my orders are right here"*
3. *"One tap to invoice — I got paid before I left the driveway"*

2. Project Classification

Technical Type	SaaS B2B — Web-first (responsive web in MVP; native mobile app Post-MVP)
Domain	Field Service Management (General)
Complexity	Medium
Project Context	Greenfield — new product
Architecture	Enterprise-ready from day one (modular monolith, feature flags, multi-tenancy)
Target Market	Solo service providers aged 30-60 with varying technical proficiency — must work for 70-year-old contractors, non-native English speakers, and workers with limited technology experience
Competitive Landscape	No direct competitors in this segment. ServiceTitan/Workiz serve teams and enterprises. Invoice Simple serves basic invoicing needs only. Vision 360 CORE fills the gap.

3. Success Criteria

This section defines measurable success indicators for user outcomes, business outcomes, and technical performance.

3.1 User Success

Peter's "worth it" moment is **financial clarity and control**:

Success Indicator	Measurable Outcome
Financial Visibility	Peter can see his revenue AND expenses in one place — he finally knows if he's making money
Professionalism	Estimates and invoices look as polished as enterprise competitors
Smart Pricing	Peter adjusts prices based on actual tracked expenses — stops leaving money on the table
Daily Clarity	Opens app → sees entire day at a glance (no hunting, no guessing)
Speed to Payment	Estimate → Invoice → Payment in under 2 minutes total interaction time

Key User Success Metrics:

- Time to create estimate: <60 seconds on-site
- Estimate-to-invoice conversion: 1 tap
- Invoice-to-payment cycle: same day (before leaving the job site)

3.2 Business Success

Timeframe	Success Metric	Target
3 Months	Active Usage	Users creating at least 1 estimate OR invoice per week
12 Months	Organic Growth	Word-of-mouth among tradespeople — measurable via “How did you hear about us?” tracking

North Star Metric: Active weekly users creating estimates — indicates product solves the core problem.

Secondary Indicators:

- Trial-to-paid conversion rate
- User retention (month-over-month)
- NPS score (“Would you recommend to a colleague?”)

3.3 Technical Success

Requirement	Target	Rationale
Reliability	99.9% uptime, zero data loss	Peter’s business depends on this — lost invoices = lost money
Responsiveness	Seamless on phone, tablet, desktop	Works in vehicle (phone), at kitchen table (tablet), home office (desktop)
Performance	App loads <3 seconds on 4G	Peter checks between jobs — no patience for slow apps
Data Integrity	All transactions persisted immediately	Invoice sent = invoice stored. No exceptions.

v2.0 note: Platform priority is now **web-first** (D-18-10) — desktop/web is the primary target (~99% of use), responsive down to tablet/phone; the native field mobile app is Post-MVP. Read the “Responsiveness” row above in that light.

3.4 Measurable Outcomes

User Outcome: “I know exactly what I earned and spent this month — and my estimates look professional.”

Business Outcome: “Users are actively creating estimates weekly and telling other tradespeople about us.”

Technical Outcome: “The app works reliably on any device, and no user has ever lost data.”

4. Product Scope

4.1 MVP — Minimum Viable Product (CORE)

Left Navigation (9 items, per approved design): Home · Schedule · Clients · Jobs · Estimates · Invoices · Payments · Expenses · Items. *Reversals vs v1.0:* “Create” moved to the top bar (D-09-2); “Reports” and “Settings” moved out of the left nav (D-09-4, D-08-15); the scheduling module is **Schedule** (not “Calendar”), and a **Home** dashboard was added (D-08-27 → #09).

Top Bar: Create (+) quick-create menu (first in row) · global Search (ServiceTitan-style) · Reports (top tab on the Home / Business Insight screen) · Notifications · Help Center · Settings (gear) · Account (user initials). *No AI Assistant / Phone-VoIP / Messaging placeholders in CORE — removed, no “coming soon” stubs (D-09-3).*

UI/UX: Collapsible left navigation (ServiceTitan-style “Collapse Menu” label) · **web-first** design — optimized for desktop/web (~99% of use), responsive down to tablet/phone (D-18-10) · **breadcrumbs** for back-navigation (D-12-3) · consistent card headers, kebab “more actions”, and pencil quick-edit across all modules.

Now in CORE (was PRO/Post-MVP in v1.0): a simple **Schedule board** — day/week/month views, up to ~3 technicians (solo + up to 2 employees, D-09-5), drag-and-drop scheduling, status icons, and a **movable/floating route map** (D-27-*). Advanced multi-technician scheduling, business units and route optimization remain PRO.

4.2 Growth Features (Post-MVP)

Feature	Description	Priority
Native Mobile Apps	iOS and Android apps (currently web-only)	High
Offline Mode	Create estimates with no signal — sync when connected	High
Autocomplete	Address and item autocomplete for faster data entry	Medium
AI Assistant	Move from placeholder to functional AI features	Medium

4.3 Vision (Future Tiers — Out of Scope)

Tier	Key Additions
PRO	Advanced multi-technician Schedule Board, Inventory, QuickBooks integration, business units, route optimization
MAX	SMS/VoIP, Payroll, Marketing automation, Checklists
Enterprise	Branch isolation, Corporate reporting, Multi-tenancy, API integrations

5. User Journeys

v2.0 note: These journeys are aspirational persona stories, retained as-is. MVP delivery is **web-first** (responsive web); the on-site/phone usage some journeys imply runs responsively in the browser, while a dedicated native field app is a **Post-MVP** growth feature (D-18-10).

5.1 User Types

User Type	Description	Interacts Via
Peter (Primary User)	Solo service operator — creates estimates, invoices, tracks payments and expenses	Vision 360 app (web responsive)
Peter (Admin)	Same user, admin role — sets up company, configures settings, manages items	Vision 360 app
Peter's Customer	Receives estimates/invoices, signs electronically, pays	Email links (no app required)

5.2 Journey 1: Peter's Morning — “My Day at a Glance”

Peter wakes up at 6:30 AM. Before Vision 360, he'd spend 20 minutes at the kitchen table piecing together his day from scattered notes, text messages, and his phone calendar. Half the time he'd forget an address or show up at the wrong time.

Now, he opens Vision 360 while drinking his coffee. Instantly, he sees:

- **9:00 AM** — Sarah Johnson, HVAC inspection, 123 Oak Street
- **2:00 PM** — Mike Thompson, water heater install, 456 Pine Ave
- **5:00 PM** — Lisa Chen, AC unit repair, 789 Elm Road

He taps Sarah's job card — her phone number, address (with map link), and notes are right there: “Prefers morning appointments. Dog's name is Max — leave gate open.”

Peter grabs his keys. He knows exactly where he's going.

Capabilities Revealed: Home screen with daily schedule, job cards with client details, tap-to-call, tap-to-navigate.

5.3 Journey 2: Peter On-Site — “Estimate in 60 Seconds”

Peter finishes inspecting Mrs. Rodriguez's AC unit. It needs a new compressor — about \$1,800 in parts and labor. Before Vision 360, he'd scribble numbers on a notepad, promise to “send something later,” and sometimes forget.

Now, standing in her living room, he taps **Create → New Estimate**. He selects Mrs. Rodriguez from his client list (her info auto-fills). He picks “AC Compressor Replacement” from his Items — the description, rate, and labor are already set.

He adds a second option: “AC Compressor + Full System Tune-Up” for \$2,100.

In 45 seconds, he shows Mrs. Rodriguez the estimate on his phone. “I can email this to you right now with both options. You sign electronically, and we’re good to go.”

She’s impressed. “That looks professional — like ServiceTitan, but you’re just one guy!”

Capabilities Revealed: Quick create menu, client selection with auto-fill, item catalog, multi-option estimates, on-device preview, email send.

5.4 Journey 3: Peter Gets Paid — “One Tap to Invoice”

Mrs. Rodriguez signed Option 2 (the tune-up bundle). Peter completed the work. Before Vision 360, he’d have to go home, type up an invoice, calculate tax, and email it — sometimes days later.

Now, standing in her driveway, he opens the signed estimate and taps **Convert to Invoice**. Vision 360 pulls in the approved option, applies tax settings, and generates a professional invoice with his logo.

He taps **Send** — the invoice goes to Mrs. Rodriguez’s email with a Stripe payment link.

His phone buzzes 3 minutes later: **Payment received — \$2,247.00**

Peter hasn’t even left the driveway.

Capabilities Revealed: Estimate-to-invoice conversion (1 tap), automatic tax calculation, branded invoice with logo, Stripe payment link, payment notification.

5.5 Journey 4: Peter’s Customer — “Easy to Sign, Easy to Pay”

Mrs. Rodriguez checks her email on her phone. She sees a professional estimate from “Peter’s HVAC Services” — company logo, itemized options, warranty information.

She taps **Option 2** and sees the full breakdown. She taps **Sign Estimate**. A popup asks: “Confirm your signature name.” She types “Elena Rodriguez” — it appears in cursive. She taps **Confirm**.

Done. No app to download. No account to create.

Two days later, Peter finishes the work. She gets an invoice email — same professional look. She taps **Pay Now**, enters her card on Stripe’s secure page, and she’s done.

She tells her neighbor: “You should call Peter — he’s got his act together.”

Capabilities Revealed: Client-facing estimate review (web), option selection, electronic signature, client-facing invoice (web), Stripe checkout integration.

5.6 Journey 5: Peter as Admin — “Setting Up His Business”

Peter just signed up for Vision 360. The welcome screen asks for his company details:

- Company name: Peter’s HVAC Services
- Email, phone, address
- Upload logo

Next, Settings:

- Tax rate (if applicable): 8.25%
- Default payment terms: Due on receipt
- Stripe account: Connected in 2 minutes

Finally, Items. Peter opens his old spreadsheet with 50 services and prices. He uses **Import** to upload them all at once. Each item now has:

- Item description (for him): “3-ton AC compressor replacement”
- Sales description (for clients): “Complete AC compressor replacement including new pad, refrigerant recharge”

Setup took 30 minutes. Now every estimate and invoice pulls from his price book automatically.

Capabilities Revealed: Company profile setup, logo upload, tax settings, payment settings, Stripe integration, item import/export, dual descriptions (internal/customer-facing).

5.7 Journey 6: Peter’s Friday — “Do I Have Money?”

It’s Friday evening. Peter’s wife asks: “How’d we do this week?”

Before Vision 360, Peter would shrug. He had no idea until tax time.

Now, he opens **Reports**. He sees:

- **Revenue this week:** \$8,450
- **Expenses this week:** \$3,200 (parts, gas, supplies)
- **Outstanding invoices:** \$1,250 (two clients haven’t paid yet)

He taps the outstanding invoices — sees Mike Thompson owes \$750 from Tuesday. He taps Mike’s contact info and calls him directly. “Hey Mike, just following up on that invoice from Tuesday...” Mike apologizes and pays via the link that same evening.

Peter tells his wife: “We cleared about \$5,000 this week after parts.” She smiles. For the first time, they actually know where they stand — and who still owes them.

Capabilities Revealed: Revenue reports, expense tracking, outstanding invoice visibility, invoice status, client contact access from invoice.

5.8 Journey Requirements Summary

Journey	Key Capabilities Required
Morning Glance	Home screen, daily jobs list, job cards, client details, tap-to-call, tap-to-navigate
Estimate On-Site	Create menu, client lookup, item catalog, multi-option estimates, preview, email send
One Tap to Invoice	Estimate→invoice conversion, tax calculation, branded invoices, Stripe link, payment notifications

Journey	Key Capabilities Required
Customer Experience	Web-based estimate review, option selection, confirmation, web invoice, Stripe checkout
Admin Setup	Company profile, logo, tax/payment settings, Stripe connect, item import/export
Financial Clarity	Reports (revenue/expenses), outstanding invoices list, invoice status tracking

6. Functional Requirements

This section details all functional requirements for Vision 360 CORE, organized by module. Feature IDs are provided for traceability to the BRD.

6.1 FR-1: Authentication & Onboarding

ID	Requirement	Priority	Acceptance Criteria
FR-1.1	Create Account — User can register with email address and password	Must Have	Email validation, password strength requirements met, terms acceptance required
FR-1.2	Login — User can authenticate with email/password	Must Have	Valid credentials grant access, “Remember me” option available
FR-1.3	Two-Factor Authentication (2FA) — Email verification code sent on login	Must Have	Code delivered within 30 seconds, “Skip on this device” option for trusted browsers
FR-1.4	Password Reset — User can request password reset via email	Must Have	Reset link expires after 24 hours, successful reset logs user in
FR-1.5	Logout — User can log out from current device	Must Have	Session terminated, user redirected to login
FR-1.6	Welcome Screen — Simple onboarding screen after first login	Should Have	Skippable, optional tip displayed, minimal steps

Note: Social login (Google, Apple, Facebook) is NOT included in CORE to minimize cost and complexity.

FR-1 — v2.0 updates (Onboarding):

- **Onboarding = 4-step flow:** (1) email + password + accept Terms; (2) email **verification code**; (3) company details — **Company name (required)**, first/last name, business phone, **industry** (predefined list + “Other”), **team size** (*Just me / Under 10 / More than 10*); (4) quick-start (Add first customer · Create first job) **or Go to Home**. (*D-09-18 company name mandatory — Marek “Yes” [41:01]; D-25-15*)

minimal registration, remaining company data added later; onboarding must never block starting to use the app.)

- Field label is **“Email”**, not “work email” — many solo operators have no work domain.
- **7-day free trial, no credit card** at signup; the credit-card / subscription step comes after onboarding (within 30 days).
- On the final onboarding step, **“Go to Dashboard”** → **“Go to Home”**, and the “Explore the sample company” option is **not placed on that step** (goal: get the user straight into the app) (D-17-20). The **sample/sandbox company is retained as a feature** — available under **Help Center**, plus rotating billboards and new-user notifications (first 30 days) (D-17-21, D-27-12).

6.2 FR-2: Company & Settings

ID	Requirement	Priority	Acceptance Criteria
FR-2.1	Company Profile — CRUD operations for company details	Must Have	Company name, email, address, phone editable; logo upload supported
FR-2.2	Logo Upload — Upload and display company logo	Must Have	Logo appears on all estimates and invoices
FR-2.3	Tax Settings — Configure tax calculation method	Must Have	Options: On Total, Deducted, Per Item, None; rate as percentage; inclusive/exclusive toggle
FR-2.4	Invoice/Estimate Defaults — Configure document defaults	Must Have	Editable: title, numbering format, default notes, currency selection
FR-2.5	Payment Settings — Configure payment processor	Must Have	Activate/deactivate Stripe; connect Stripe account via OAuth

FR-2 — v2.0 updates (Company & Settings):

- All settings in **one place**, organized as an **accordian** (expandable categories) that scales as more are added (D-11-1).
- **FR-2.3 Tax → Tax Profiles:** replace “On Total / Deducted / Per Item / None” with a **Tax Profile** dropdown; profiles are created/edited in Settings (renamed from “tax groups”) (D-08-23, D-10-3, D-15-2).
- **FR-2.4 → drop custom numbering:** invoice/estimate numbering is **system-generated** (customer-ID + sequence, e.g. 10245-J01), not a user-defined format (#13 — “no value in this feature”). Default notes + Terms & Conditions live in Settings (thank-you note per D-14-3; T&C per D-24-12), not per-document.
- **Regional** (country, time zone, date/time format, **week starts Sunday**), **business hours**, editable **Lead sources**, **Vendors** list, **notifications** list (estimate-heavy), **language** selector — EN · ES · PT · FR · DE · DA · NL (D-25-18).
- **Appearance = navigation only** (sidebar *Expanded / Collapsed*). **Theme (light/dark) is NOT in MVP — removed** (D-25-17).

- **Billing & Plan:** subscription details, account-manager contact, payment method (**Stripe**), cancel option, payment history; “more advanced plans coming soon”.

6.2b FR-2b: Team, Roles & Permissions (RBAC) *(new in v2.0)*

Predefined roles: Super Admin (owner) · Admin · Worker. Admin (e.g. an office assistant) can edit Workers’ role-based access; Super Admin is highest (D-19-16 — “super admin and admin” concept approved).

Custom roles: clone a preset → rename → adjust. Permissions are **per-module** (*View / Create / Edit / Delete / None*) and are **attached to the individual user** — the role is a customizable **preset** you copy from (D-15-7). MVP ships a practical subset (~24 permissions), not the full enterprise set (~186).

Manage Team: invite user (email → user sets password); user statuses **Active / Inactive / Invited** (Invited = from invite until first workspace login) (D-29-16); pay-rate settings (salary / per day / per hour); login & security (temp-password link on invite **or** force password change on first login; 2FA via email/phone); manage team custom fields.

Financial visibility gate: job financials (profit margin, costs) can be hidden from technicians/dispatchers/CSRs via RBAC on/off (mostly Pro) (D-29-12).

6.3 FR-3: Client Management

ID	Requirement	Priority	Acceptance Criteria
FR-3.1	CRUD Client — Create, read, update, delete client records	Must Have	Fields: name, email, address, phone, custom notes
FR-3.2	Client List — View all clients with search and filter	Must Have	Searchable by name/phone/email; sortable; shows total billed
FR-3.3	Client Detail View — View complete client profile	Must Have	Shows all client info plus work summary (invoices, estimates, jobs)
FR-3.4	Quick Actions — Create job/estimate/ invoice from client profile	Must Have	One-tap creation with client auto-populated
FR-3.5	Address Autocomplete — Auto-suggest addresses during entry	Should Have	Google Places or any other relevant service integration for quick address entry
FR-3.6	Deactivate Client — Mark client as inactive	Should Have	Inactive clients hidden from default views but data retained

FR-3 — v2.0 updates (Clients):

- **FR-3.1 → no hard delete:** clients are **Deactivated** (hidden from default view, data retained) — never deleted (D-08-16, D-14-19).

- **Statuses: Active / Prospect / Inactive.** List filter *Active / Inactive / All*; **Prospects KPI card**; software auto-flips **Prospect → Active on first payment** (not manually) (D-20-1).
- **Residential only** in the create form — the residential/commercial *type* is dropped from MVP; the commercial icon + reporting split are Pro (D-08-12).
- **2 custom fields** per client (defined in Settings); client **tags** are **Pro** (D-08-7, D-10-16).
- **Additional contacts** with a **relationship dropdown** (daughter, son, tenant, property manager, caregiver, HOA rep, ...); **secondary phone + extension**; **middle initial**; **preferred name** (D-17-14, D-03-*).
- **Addresses:** billing default + “**service same as billing**” checkbox; **multiple properties** supported (service-only); one marked **Primary** shows on the profile (D-08-9, D-10-9, D-14-22).
- **Manage duplicates: Merge** (older customer-ID survives; bulk merge for large sets) + “**Not a duplicate**”; jobs have no duplicates (D-20-1/5/8/11).
- **Account statement** (client-facing): columns date · job · invoice · payment, color-coded, **must include jobs**; send / print (D-20-16).
- **Collect payment & Send payment reminder** (bulk, filter by balance) from the client level; SMS reminders = Post-MVP, **email only** (D-20-24, D-20-25).
- **FR-3.5 Address Autocomplete** (Google Places / equivalent) — retained (Should Have).

6.4 FR-4: Items / Services Catalog

ID	Requirement	Priority	Acceptance Criteria
FR-4.1	CRUD Item — Create, read, update, delete catalog items	Must Have	Fields: item description (internal), sales description (customer-facing), rate, taxable toggle
FR-4.2	Items List — View all items with search and filter	Must Have	Searchable, sortable, filterable by category
FR-4.3	Item Import/Export — Bulk import/export items via CSV	Must Have	CSV template provided; import validates data before committing
FR-4.4	Dual Descriptions — Separate internal and customer-facing descriptions	Must Have	Item description for operator; sales description shown to customer

FR-4 — v2.0 updates (Items / Services Catalog):

- **Item types = 5, predefined:** Service · Material · Equipment · Asset · **Admin** (the “Fees” type was renamed **Admin** — non-sellable: discounts, adjustments, warranty, late/finance charges) (D-26-8, D-30-12). Price book is **not** a type — it is a customer-facing view of items.
- **Categories** fully customizable per industry; the item form uses a **single Type + Category dropdown**; **no subcategory** in MVP (D-26-8, D-26-10, D-28-17).

- **Price book vs All-items:** price book = customer-facing item at a **flat rate** — **not composed of multiple sub-items / part prices** (no cost / margin / taxable columns shown) (D-28-15); All-items = full internal list with tabs (Service / Material / Equipment / Assets / Fees→Admin).
- **Item fields:** user-entered **item code**; **SKU not in MVP** (D-29-10); price vs cost; taxable; **2 custom fields**; **notes** (for rare one-off items); default quantity; tax profile; manufacturer / vendor / department dropdowns (D-26-9, D-26-11, D-26-13).
- **Bulk actions:** **Deactivate** + Download CSV; hard **delete allowed only for unused / trash items** — an item used on a job cannot be deleted, only deactivated (D-26-12, D-28-14).
- **CSV import/export** for items and **vendors** (vendor = name / code / contact, editable dropdown like counties) (D-01-14, D-18-11).
- ★ **Recurring item charging:** the per-item **“charge per appointment / once / don’t charge”** menu is **NOT in MVP** — removed; prepaid maintenance handled via a **\$0 prepaid line item + a separate charged item** (D-29-4, ← part of D-26-17).

6.5 FR-5: Estimates

ID	Requirement	Priority	Acceptance Criteria
FR-5.1	Create Estimate — Build estimate with options	Must Have	Auto-populate company info; select client; add up to 4 options
FR-5.2	Estimate Options — Support multiple pricing alternatives	Must Have	Each option contains items with descriptions, rates, quantities
FR-5.3	Add Items to Options — Select from catalog or enter manually	Must Have	Items auto-populate from catalog; manual entry supported
FR-5.4	Warranty & Notes — Add warranty info and additional notes	Should Have	Free-text fields displayed on estimate
FR-5.5	Preview Estimate — View estimate as customer will see it	Must Have	Mobile-friendly preview renders professional layout
FR-5.6	Send Estimate — Email estimate to customer	Must Have	Email contains link to web-based estimate review
FR-5.7	Request Signature — Send estimate for electronic signature	Must Have	Customer receives email with sign link
FR-5.8	Convert to Invoice — One-tap conversion of approved estimate	Must Have	Approved option transfers to invoice with all details
FR-5.9	Create Job from Estimate — Generate job from estimate	Should Have	Job created with estimate details and client info

ID	Requirement	Priority	Acceptance Criteria
FR-5.10	Export to PDF — Generate PDF of estimate	Must Have	Professional PDF generated for download/print
FR-5.11	Estimates List — View all estimates with filters	Must Have	Filter by status (Draft, Sent, Signed); search; sort

FR-5 — v2.0 updates (Estimates):

- **Options: 3 options = 3 tabs, IN MVP** (option-sheet good/better/best) — supersedes v1.0 “up to 4 options” and the earlier “single option for MVP” (D-31-13 ← D-13-7, D-22-12).
- **Statuses:** Draft · Sent · Viewed · Approved · Rejected · Expired · Archived. **Expired = timer only.** The job’s estimate-selector **excludes** Draft / Rejected / Archived and **includes** Expired (D-22-10).
- ★ **Expiration default = 30 days, configurable in Settings** (was same-day-as-creation) (D-25-1 ← D-24-3).
- **Templates:** selectable pre-built templates, **two sizes — Letter (8.5×11) & Legal (8.5×14)**; color/layout only (fields cannot be moved).
- **Deposit** tab (percentage **or** fixed amount; “due upon” approval / scheduling / start), shown when a deposit is required.
- **Notes:** *client notes vs internal/private notes*; the default **thank-you note moved to Settings** (D-14-3); **Terms & Conditions render as full pages inside the PDF** + a page-1 disclaimer + a link in the covering email (D-24-12).
- **Convert to job(s):** one estimate → **multiple jobs** (line items copied into each) (D-22-8); the estimate detail has **no “add jobs” section** — just header + line items + a related-jobs tab (D-24-7).
- **Archive**, not delete, for estimates.

6.6 FR-6: Estimate Signing (Client-Facing)

ID	Requirement	Priority	Acceptance Criteria
FR-6.1	View Estimate — Customer views estimate via web link	Must Have	No app download required; mobile-responsive layout
FR-6.2	Select Option — Customer selects one option from estimate	Must Have	Clear option presentation; selection confirmation
FR-6.3	Electronic Signature — Customer signs by typing name	Must Have	Name rendered in cursive font; timestamp captured
FR-6.4	Confirmation — Customer receives confirmation email	Must Have	Email confirms selection and signature
FR-6.5	Signed Estimate Display — Signature and date shown on final document	Must Have	Professional display of signature block

FR-6 — v2.0 updates (Estimate Signing):

- **Signature settings** live in **Settings** (configured per estimate/invoice, not editable by technicians). Require customer signature **before start of work** and **on completion**; the “**parent/guardian signature when a minor is present**” option is dropped (D-31-9).
- Client-facing web review retained: no app download / no account, clear option selection, type-name (cursive) signature with timestamp, confirmation email.

6.7 FR-7: Invoices

ID	Requirement	Priority	Acceptance Criteria
FR-7.1	Invoice Structure — Standard invoice layout	Must Have	Logo, company info, client info, number/date, terms, items, total, notes
FR-7.2	Create Invoice — Build invoice manually or from estimate	Must Have	Auto-populate from estimate or create from scratch
FR-7.3	Item Sections — Add line items with details	Must Have	Description, rate, quantity, amount, tax checkbox; select from catalog
FR-7.4	Discount — Apply discount to invoice	Should Have	Percentage or flat amount discount
FR-7.5	Payment Terms — Set payment terms	Must Have	Options: Due on receipt, Net 15, Net 30, custom
FR-7.6	Preview Invoice — View invoice as customer will see it	Must Have	Mobile-friendly preview with payment button
FR-7.7	Send Invoice — Email invoice with payment link	Must Have	Email contains invoice and Stripe payment link
FR-7.8	Export to PDF — Generate PDF of invoice	Must Have	Professional PDF for download/print
FR-7.9	Print Invoice — Print directly from app	Should Have	Print-optimized stylesheet
FR-7.10	Duplicate Invoice — Copy existing invoice	Should Have	Creates draft copy for editing
FR-7.11	Invoices List — View all invoices with filters	Must Have	Filter by status (Outstanding, Paid); search; sort

FR-7 — v2.0 updates (Invoices):

- **Statuses (MVP): Unpaid · Overdue · Paid · Partially paid · Void** — no “deposited / not-deposited” (those need bank reconciliation → Post-MVP) (D-16-15).
- **Line items are NOT mandatory** on an invoice (D-29-14). The invoice is the order to pay.

- **Multiple jobs on one invoice:** job details shown as an **accordion** in place (expand/collapse), not a separate tab; with a single job, shown inline (D-24-16).
- **Batch invoicing:** many jobs (e.g. for one investor / property-management customer) → **one invoice** (select multiple jobs), so payment is collected once.
- **Void and Archive** (not hard delete) are the retention actions for invoices — a voided invoice is non-editable (D-18-3).
- **Numbering** system-generated (customer-ID + sequence, e.g. 10245-J01); no custom numbering.

6.8 FR-8: Payments

ID	Requirement	Priority	Acceptance Criteria
FR-8.1	Accept Online Payments — Generate Stripe payment link	Must Have	Link included in invoice email
FR-8.2	Record Manual Payments — Log cash/check payments	Must Have	Record date, method, amount, reference
FR-8.3	Partial Payments — Accept deposits and partial payments	Must Have	Track remaining balance; payment history per invoice
FR-8.4	Payment Status Tracking — View payment status	Must Have	Status: Pending, Partial, Paid; track by customer/job
FR-8.5	Payments List — View all payments with filters	Must Have	Filter by amount, date, status, method, customer
FR-8.6	Payment Notification — Push notification on payment received	Must Have	Notification within 60 seconds of payment
FR-8.7	Export Payments — Export payment data to CSV	Should Have	All payment records exportable

Supported Payment Methods:

- Cash (manual record)
- Check (manual record)
- Credit/Debit Card (Stripe integration)
- Deposits/Partial payments

FR-8 — v2.0 updates (Payments):

- **Collect vs Record:** **Collect payment** = integrated **credit card** (Stripe) — actually processed/charged; **Record payment** = external methods (cash, check, Zelle, Venmo, Cash App, bank transfer, financing) captured with **date + reference number only** (no integration) (D-24-28, D-20-21).

- **Payment always starts from an invoice.** The Collect-payment form lists the customer's open invoices — a fully-paid invoice drops off the list (D-31-4). **One payment can settle several invoices when each is paid in full** (select them; amounts sum — Marek [#31 49:56]: *"we will add one more invoice, let's say the second one"*). **When a payment is NOT paying in full** (a partial / deposit that can't fully cover the selected invoices) → **one invoice at a time** (one payment ↔ one invoice), to keep allocation simple (Marek [#31 53:23]: *"If they are not paying in full. One invoice at the time"*). This refines the earlier blanket one-invoice rule (D-25-7) via the newest source #31 (D-31-4/5).
- **Partially-paid** status comes straight from the invoice; a **fully-paid invoice does not appear** in the payment's invoice list (D-31-4).
- **Preferred payment method** stored at the **client level**; the Collect-payment form pre-selects it (card → collect/charge; cash/check/etc. → record + reference #).
- **Refund** (not delete) is the reversal action; MVP keeps **refund only**.
- **Collect payment** is reachable from the **client** (generic — pick customer → invoice), the **invoice** (pre-filled), and the top-bar (+).

6.9 FR-9: Jobs & Work Orders

ID	Requirement	Priority	Acceptance Criteria
FR-9.1	CRUD Jobs — Create, read, update, delete jobs	Must Have	Fields: description, client, address, start/end datetime, notes
FR-9.2	Job Photos — Attach before/after photos	Must Have	Photos linked to job; accessible from job detail
FR-9.3	Job Status — Track job status	Must Have	Status: Scheduled, In Progress, Completed; manual status change
FR-9.4	Link to Estimate/Invoice — Associate job with documents	Must Have	Job links to related estimate and invoice
FR-9.5	Jobs List — View all jobs with search and filter	Must Have	Search, filter by status, sort by date
FR-9.6	Job Detail — View complete job information	Must Have	All job fields plus linked documents and photos

FR-9 — v2.0 updates (Jobs & Work Orders):

- **Statuses (July 6 call, #31) — full set: Unscheduled · Scheduled · Dispatched · In Progress · Paused · Completed · Canceled.**
- **System** statuses **Scheduled / In Progress / Completed** are fixed out-of-the-box and never change; the company can **add custom** statuses (e.g. Canceled, Paused) (D-31-11).
- **Unscheduled** (status) is applied **automatically** only when a job has **no date**; you **cannot manually switch a job to Unscheduled** (it is the default zeroth state) (D-22-5).
- **Dispatched** = the job has been pushed to the technician (added for the mobile app) (D-21-26). **Paused** = time-clock only (D-22-4).

- Trigger flow: **Unscheduled** → **Scheduled** → **Dispatched** → **In Progress** → **(Paused)** → **Completed** (on customer signature) / **Canceled** (D-31-12).
- **Pending-jobs side column** works on **chips / filters** — a layer *separate from statuses*: **Unassigned** = no technician; **Unscheduled** = no date. So a job that has a date but no technician carries the **Scheduled** status + **an Unassigned chip** (D-22-5). Pending-panel filters: *Show unassigned / unscheduled / scheduled* (D-22-2).
- **Create — mandatory**: customer required; **assigned-to NOT mandatory** (goes to the unassigned bucket); **start-time NOT mandatory** (default length comes from job type); status is chosen on create (D-19-12, D-20-26, D-23-20, D-22-13).
- **Job frequency** (one-off / recurring) — renamed from “job type”; **job type** now = work category (estimate / installation / maintenance / inspection / service) (D-30-11). **Recurring** = Google-Calendar-style engine, ends after N occurrences or a date, created as **N separate job records** (D-16-21, D-23-22, D-29-3); a one-off can be converted to recurring (D-16-22).
- **One estimate** → **many jobs**; one job → many estimates; the estimate selector excludes Draft/Rejected/Archived, includes Expired (D-22-8, D-22-10).
- **Duplicate = copy**; **Cancel** (no separate Archive) (D-22-11); **Completed = read-only/frozen** (manager/RBAC can still edit) (D-28-10); job number = customer-ID + Jnn (D-07-6, D-14-24). **No hard delete**.
- Three note types: **field** / **office** / **internal** (renamed from “private”) (D-17-5).

6.10 FR-10: Schedule

ID	Requirement	Priority	Acceptance Criteria
FR-10.1	Calendar View — Visual calendar showing jobs	Must Have	Month/week/day views; dates with jobs visually marked
FR-10.2	Daily Schedule Snapshot — View today’s jobs at a glance	Must Have	Time, customer, address for each job; primary home screen feature
FR-10.3	Job Quick Access — Tap job to view details	Must Have	Direct navigation to job detail from calendar
FR-10.4	Filter Views — Filter calendar by status/type	Should Have	Filter active jobs by week/month/status
FR-10.5	Visual Status Indicators — Color-coded job statuses	Should Have	Visual differentiation for scheduled, in-progress, completed

FR-10 — v2.0 updates (Schedule):

- Module & nav item **renamed “Calendar” → “Schedule”**; a **“Home”** dashboard (Business Insight) was added to the nav (D-08-27 → #09).
- **Schedule board in CORE**: day / week / month; up to ~3 technicians (rows) — solo + up to 2 employees (D-09-5); **drag-and-drop**; dropping a job from Pending onto the board sets its duration from the job type.
- **Compact rows** (2 lines: *customer + city / job name + status*); **KPI blocks removed** from the board (D-27-1, D-27-3).

- **Status as an icon** (calendar = scheduled, ►/wrench = in progress, ✓ = completed) rather than words; **background colour = job type**; a **\$ sign** marks a job that has a sold amount (>\$0) instead of showing full revenue (D-27-6, D-27-7, D-27-8).
- **Route map: floating / movable / openable in a new tab / toggled on-off** (not embedded in the board); clicking a job highlights its pin on the map (D-27-9, D-27-10).
- Pending column = “Pending jobs” (Unassigned / Unscheduled) — see FR-9.
- Advanced multi-technician board, business units and route optimization remain **PRO**.

6.11 FR-11: Expenses

ID	Requirement	Priority	Acceptance Criteria
FR-11.1	Record Expense — Manual expense entry	Must Have	Fields: merchant, category, date, total, notes
FR-11.2	Attach Receipt — Add photo of receipt	Must Have	Photo stored with expense record
FR-11.3	Scan Receipt — Capture receipt via camera	Should Have	Camera integration for quick capture
FR-11.4	Link Expense — Associate expense with job/invoice	Should Have	Expenses linked to job auto-link to subsequent invoice
FR-11.5	Expenses List — View all expenses with filters	Must Have	Filter by category, date, job; search
FR-11.6	Profit Calculation — Calculate profit per job	Should Have	Revenue minus linked expenses

FR-11 — v2.0 updates (Expenses):

- **Expense-job link (optional, two categories):**
 - A **job-specific** expense — materials, labor, equipment for *that* job — is **linked to the job** → job costing / profit-per-job (D-25-8, D-30).
 - **Shared / overhead** expenses — office rent, insurance, marketing, and anything bought once for several jobs (e.g. a **fuel canister**) — are **not linked to any job**: they are recorded with a date and surface in the **company dashboard** (business revenue-vs-expenses), not in per-job costing (D-29-15; Marek [#29 01:09:19]: “office rent... you do not ring to the job”).
- Expenses **never** link to invoices — the old invoice link was removed (Marek [#25 39:37]: “The expenses are linked to the job, not invoices”).
- *Resolved on double-check (Solomiia)*: job-specific → linked (job costing); shared/overhead incl. cross-job fuel → unlinked → dashboard. Consistent with #25 / #29 / #30.
- **Vendor** = dropdown, editable in Settings (like counties); “**add another expense**” from the job level pre-populates vendor / job / category for fast entry of a stack of receipts (D-16-16, D-16-17).
- **Invoice-number field** on an expense = the **vendor’s receipt/invoice number** (distinct from linking to a Vision360 invoice) (D-16-18).

- **Archive** normally; **delete allowed** for a returned purchase (to restore the job's profit) (D-25-14).
- **No tax** on expense line items (US model — expense stored as one total incl. tax) (D-29-1).
- Company overhead (rent, fuel, phone, marketing, insurance) recorded with a date and **never linked to jobs** (D-29-15).
- Attach receipt photo / camera scan; profit per job = revenue – linked expenses.

6.12 FR-12: Reports (A preliminary description. The requirements have not yet been finalized)

ID	Requirement	Priority	Acceptance Criteria
FR-12.1	Revenue/Expense Report — View financial summary	Must Have	Per period: week, month, year; revenue, expenses, profit
FR-12.2	Client Report — Revenue breakdown by client	Should Have	Total invoiced, amount paid per client
FR-12.3	Item Report — Performance by catalog item	Should Have	Quantity sold, revenue per item
FR-12.4	Outstanding Invoices — View unpaid invoices	Must Have	List of outstanding invoices with amounts and client info
FR-12.5	Export Reports — Export reports to PDF	Should Have	Professional PDF for accounting/records

FR-12 — v2.0 updates (Reports):

- **Concept approved** (D-31-6): categories (Financial/Business; Clients, Jobs, Estimates; Team Operations), **quick filters + custom range**, column select, **export to Excel / generate / print**.
- **Generated reports** listed as a **table under the report**, and stored in the **activity tab** (which also shows scheduled-send history) (D-31-7).
- **Advanced-filter “bubbles” included in Generate report**, so a scheduled report shows exactly which data it covers (D-31-8).
- **Scheduled email reports are IN scope** (weekly / daily / monthly to recipients) — replaces printing (D-30-4, ← D-28-2 “not for MVP”).
- **Renames: “Sales report”** (was “Revenue report”) (D-30-2); **“Invoice AR aging”** (was “Invoice summary”), defaulting to unpaid / past-due with clickable aging buckets (D-30-7).
- **Scope:** design only the important “rock/gravel” reports for MVP; “sand” reports not designed (D-28-4). Per-user report access (D-29-17).
- **Sales tax report** — requirements still pending (Marek's homework, #31 [54:23]) → **TBD**.

6.13 FR-13: Notifications

ID	Requirement	Priority	Acceptance Criteria
FR-13.1	Payment Notification — Alert when invoice is paid	Must Have	Push notification (on mobile) in-app notification (on web) on payment confirmation
FR-13.2	Estimate Signed Notification — Alert when estimate is signed	Must Have	Push notification with option selected

FR-13 — v2.0 updates (Notifications):

- Notification list is **estimate-heavy**: estimate sent / viewed / changes requested / approved; client signs estimate or invoice; payment received; invoice overdue.
- Delivery for MVP = **email + in-app**; **SMS is Post-MVP** (D-20-25); quiet-hours = Pro.
- Notifications configurable per user (and company defaults in Settings).

6.14 FR-14: Global Search

ID	Requirement	Priority	Acceptance Criteria
FR-14.1	Unified Search — Search across all data types	Must Have	Search customers, jobs, invoices, estimates by any field
FR-14.2	Search Filters — Scope search to specific data types	Should Have	Filter search to Customers, Jobs, Invoices, Estimates

FR-14 — v2.0 updates (Search):

- Two levels: a **global** top-bar search (across Clients / Jobs / Invoices / Estimates) **and** a **local per-page search** on each list page (D-06). Global search offers scope filters by data type; recent results shown.

7. UI/UX Requirements

Screen Inventory (CORE)

Screen	Primary Function
Login	User authentication
Registration	New account creation
Verify	2FA email verification
Reset Password	Password recovery
Welcome	Optional onboarding (skippable)
Home	Daily schedule snapshot + primary navigation
Company	Business details and logo management
Settings	Tax, invoice/estimate defaults, payment options
Clients	Client list with search/filter

Screen	Primary Function
Client Detail	Individual client + work summary
Items	Item/service catalog
Item	Individual item details
Estimates	Estimate list with filters
Estimate	Create/edit estimate with options
Invoices	Invoice list with filters
Invoice	Create/edit invoice
Payments	Payment list and tracking
Jobs	Job list and history
Job	Individual job details + photos
Calendar	Calendar view with jobs
Expenses	Expense list
Expense	Individual expense entry
Reports	Revenue, expense, profit reports

v2.0 Screen Inventory updates: the “Calendar” screen is now **“Schedule”** (schedule board + route map); **Home** (Business Insight) is the landing screen with **Reports** as a top-tab; Settings is reached via the top-bar gear. Bottom-navigation / hamburger screens are Post-MVP (native app).

Navigation Architecture

Left Navigation (9 items — per approved design): Home · Schedule · Clients · Jobs · Estimates · Invoices · Payments · Expenses · Items. *(Create moved to the top bar — D-09-2; Reports & Settings moved out of the left nav — D-09-4, D-08-15; “Calendar” renamed “Schedule” and “Home” added — D-08-27 → #09.)*

Top Bar: Create (+) quick-create (first in row) · global Search (ServiceTitan-style) · **Reports** (Home / Business Insight top-tab) · Notifications · Help Center · **Settings** (gear) · Account (user initials). *No AI Assistant / Phone-VoIP / Messaging placeholders in CORE (D-09-3).*

Navigation Behavior: collapsible left navigation (ServiceTitan-style “Collapse Menu” label); **breadcrumbs** for back-navigation (D-12-3); consistent card headers + kebab “more actions” + pencil quick-edit across modules. *Bottom-navigation / hamburger layouts belong to the **Post-MVP native mobile app**, not the web-first CORE.*

Design Principles

Principle	Implementation
Web-First	Design for desktop/web first (~99% of use); responsive down to tablet/phone; native field app Post-MVP (D-18-10)
Simplicity Mandate	“Design for six year olds” — click, click, click, send

Principle	Implementation
Steve Test	70-year-old contractor must complete estimate in <15 min without training
Duolingo Model	Select → Execute → Done
Thumb-Friendly	Minimum 44px tap targets for all interactive elements
Professional Output	Estimates and invoices look as polished as enterprise competitors

Responsive Breakpoints

v2.0: web-first — **desktop is the primary target**; tablet/phone remain responsive (D-18-10).

Device	Minimum Resolution	Notes
Desktop	1366×768 px	Full left navigation visible
Tablet	768×1024 px	Collapsible navigation
Mobile	375×667 px	Bottom navigation, hamburger menu

Anti-Patterns to Avoid

- Over-complicated menus (hamburger navigation with deep nesting)
- Multi-step processes for simple tasks (5+ clicks to create estimate)
- Features that can't be used easily with one hand
- Tiny tap targets that require precision
- Horizontal scrolling at supported resolutions

8. Non-Functional Requirements

NFR-1: Performance

Metric	Target	Rationale
API Response Time	< 300ms average	Peter checks between jobs — no patience for slow apps
App Load Time	< 3 seconds on 4G	Quick access from vehicle
Estimate Creation	< 60 seconds	Core value proposition
Invoice Generation	< 5 seconds	One-tap conversion must be instant

NFR-2: Availability & Reliability

Requirement	Target
Uptime	99.9% per month

Requirement	Target
Data Loss	Zero tolerance — all transactions persisted immediately
Failover	Automatic failover for critical services
Hosting	Cloud-based with autoscaling

NFR-3: Scalability

- Database designed from enterprise level down (expose fields progressively)
- Architecture supports scaling from CORE → PRO → MAX → Enterprise
- No costly restructuring when adding features for larger tiers
- “No-archiving” data strategy — indefinite growth without purging

NFR-4: Security

Area	Requirement
Authentication	Email + Password; 2FA via email; “Skip on trusted device” option
Payment Processing	Handled entirely through Stripe; PCI compliance via payment gateway
Data Storage	No credit card data stored in Vision 360
Encryption	Industry-standard encryption for data at rest and in transit
API Security	Secure API endpoints with proper authentication

NFR-5: Supported Platforms

Web Application (CORE):

- Browsers: Chrome, Edge, Safari, Firefox (latest 2 versions each)
- Responsive: Desktop, Tablet, Mobile layouts

Future Native Mobile App:

- Android 9.0+
- iOS 14+
- Minimum resolution: 720×1280

NFR-6: Languages

- **Multi-language (v2.0):** English · Spanish · Portuguese · French · German · Danish · Dutch — language selector in Settings (D-25-18).
- Out of scope: Chinese, Arabic.
- *Supersedes v1.0 (“English/Spanish; US-market only; no international expansion for 3-4 years”).*

NFR-7: Accessibility

- Touch-friendly controls on all viewports
- High visibility in direct sunlight for field use

- No horizontal scrolling at supported resolutions
- Consistent navigation patterns across breakpoints

9. Data Requirements

Core Entities

Entity	Key Attributes
User	email, password_hash, 2fa_enabled, company_id
Company	name, email, address, phone, logo_url, tax_settings, stripe_account_id
Client	name, email, address, phone, notes, status, company_id
Item	item_description, sales_description, rate, taxable, category, company_id
Estimate	number, date, client_id, status, options[], notes, warranty_info, signature_data
EstimateOption	name, items[], subtotal, taxes, total
Invoice	number, date, client_id, job_id, estimate_id, status, items[], subtotal, discount, taxes, total, terms
Payment	invoice_id, amount, method, date, reference, stripe_payment_id
Job	description, client_id, address, start_datetime, end_datetime, status, notes, photos[], estimate_id, invoice_id
Expense	merchant, category, date, total, notes, receipt_url, job_id

v2.0 — Data-model updates (per DECISIONS.md):

- **Role** (name, is_system, permissions[]) + **RolePermission** (module, action ∈ View/Create/Edit/Delete/None) — predefined Super Admin / Admin / Worker + custom roles (D-19-16). **User** gains role_id, status ∈ Active/Inactive/Invited, pay_rate (D-29-16).
- **Client** gains status ∈ Active/Prospect/Inactive, is_residential, custom_field_1/2, lead_source; child **Contact**(s) (name, phone, ext, relationship) and **Property**(ies) (address, is_primary) for multiple service addresses (D-20-1, D-08-7/9/12, D-17-14).
- **Company/Tax**: TaxProfile (name, rate(s), jurisdiction) replaces flat tax fields (D-08-23, D-15-2); **Vendor** (name, code, contact) for expenses — settings-managed dropdown, editable codes (D-16-17, D-18-11).
- **Job** gains frequency ∈ one-off/recurring, job_type (work category), status ∈ Unscheduled/Scheduled/Dispatched/In Progress/Paused/Completed/Canceled (system + custom), nullable assignee_id; recurring series → N Job records (D-31-11/12, D-30-11, D-29-3). **Appointment** table (job_id, ...) seeded now, hidden in MVP (D-29-6).
- **Estimate**: options[] (up to 3 tabs), expiration_date (default +30 days), status ∈ Draft/Sent/Viewed/Approved/Rejected/Expired/Archived (D-31-13, D-25-1).
- **Expense**: nullable job_id (job-costing or overhead — **open, see FR-11**), vendor_id, document_number (vendor's receipt #); **no invoice_id** (D-25-8).

- **Report / ReportSchedule** (category, filters/"bubbles", columns, schedule ∈ none/daily/weekly/monthly, recipients[]) (D-31-6/7/8, D-30-4).
- **Cardinality fixes:** Estimate 1 → **N Jobs** (D-22-8); Job 1 → **N Invoices** (multi-trade); **Invoice N ↔ N Payments** — one payment can settle several invoices paid in full, and an invoice can receive several partial payments; a *partial* payment maps to a single invoice (D-31-4/5, refines D-25-7).

Data Relationships

Company (1) — (N) Users
 Company (1) — (N) Clients
 Company (1) — (N) Items
 Client (1) — (N) Jobs
 Client (1) — (N) Estimates
 Client (1) — (N) Invoices
 Estimate (1) — (0..1) Invoice (converted)
 Estimate (1) — (0..1) Job
 Job (1) — (0..1) Invoice
 Job (1) — (N) Expenses
 Invoice (1) — (N) Payments

10. Integration Requirements

INT-1: Stripe Payment Integration

Requirement	Details
OAuth Connection	User connects Stripe account via OAuth flow
Payment Links	Generate Stripe payment links for invoices
Webhook Handling	Receive payment confirmation webhooks
Refunds	Support refund initiation (future consideration)

v2.0 (INT-1): payment actions split into **Collect** (credit card — actually processed via Stripe) vs **Record** (external methods: cash / check / Zelle / Venmo / Cash App / bank transfer / financing — logged with date + reference #, no integration) (D-24-28). **Refund is supported in MVP** (the reversal action for a payment), not just a “future consideration”. Stripe also processes the app’s own subscription billing.

INT-2: Email Services

Requirement	Details
Transactional Email	Send estimates, invoices, confirmations
Templates	Professional email templates matching branding
Delivery Tracking	Track email delivery status

INT-3: Google Places API (Address Autocomplete)

Requirement	Details
Address Autocomplete	Suggest addresses during client/job entry
Address Validation	Validate entered addresses

INT-4: Push Notifications (for mobile)

Requirement	Details
In-app notification	Browser notifications for payments and signatures
Mobile Push	Native push for future mobile app

INT-5: Hosting & Infrastructure (Azure) (v2.0 — from #31, 2026-07-06)

- Hosted on the **client's Azure subscription** (Reenbit granted Owner access) (D-31-1).
- Stack: **Azure App Service** (back-end), **Key Vault** (secrets — DB connection string, Google Maps key), **Azure SQL Server + Database**, **Storage Account** (static front-end + files: photos / receipts / PDFs), **Application Insights + Log Analytics** (monitoring). Dev resource group Vision360-Dev, cheapest tiers for testing.
- Unused ~\$100/mo App Service plan stopped to reduce cost (D-31-2).

11. Constraints

Constraint	Description
C-1	CORE version targets solo operators ONLY — no multi-user features
C-2	Responsive web app is primary interface; native mobile deferred 8-12 weeks
C-3	US market only — no international expansion for 3-4 years
C-4	No social login in CORE (deferred to PRO+)
C-5	Payment processing via third-party only (Stripe) — no direct card handling
C-6	All hosting cloud-based — no on-premise installations

12. Assumptions

ID	Assumption
A-1	Version 1 (CORE) targets solo service operators ONLY
A-2	Responsive web application is the primary interface for CORE
A-3	Third-party payment processor (Stripe) will be used for online payments
A-4	Invoice Simple's tax module design will be used as reference

ID	Assumption
A-5	UAT will be conducted with real solo operators before major development
A-6	Database structure will be designed from enterprise level down
A-7	Communication with customers uses native share functionality
A-8	Client will provide test data and access to potential end users
A-9	Third-party services will be available via documented APIs
A-10	System designed to comply only with US-based regulations

v2.0 — Constraints & Assumptions updates:

- **C-1 / A-1 revised:** CORE now supports **multiple users + RBAC** (Super Admin / Admin / Worker + custom roles), not solo-only (D-19-16). Onboarding team-size buckets: *solo operator / under 10 / more than 10* (D-17-19).
- **C-3 / A-10 revised:** product is **multi-language** (EN/ES/PT/FR/DE/DA/NL) for more than the US market — not “US-only for 3-4 years” (D-25-18).
- **A-4:** tax is modelled as **Tax Profiles** (D-08-23, D-15-2), not a flat Invoice-Simple rate.
- **C-2 / A-2 reinforced:** web-first; native mobile deferred (D-18-10). **C-5** (Stripe-only, no direct card handling) stands.

Appendix A: Payment Scenarios

Supported Payment Methods (CORE)

Method	Timing	Location	System Support
Cash	On-site, immediate	Customer location	Manual “mark as paid”
Check	On-site or pickup	Customer location / mailed	Manual “mark as paid”
Credit Card	On-site, phone, or link	Any	Payment link (Stripe)
Deposit	Before job start	Any	Record partial payment

Extended Scenarios

- **Prepayment/Deposit:** 20-40% deposit for jobs \$10,000+; system tracks partial payment
- **Delayed Payment:** Customer may withhold until permit inspection passes; system tracks status over time

Not in CORE (Future)

- Third-party financing (Synchrony, GreenSky)
- **Automated / scheduled** payment reminders (*manual email payment reminders are in CORE — D-20-24; SMS reminders Post-MVP — D-20-25*)

- Auto-charging recurring **payments / subscriptions** (*recurring **jobs** are in CORE — D-16-21; auto-billing of recurring charges is Future*)

Appendix B: Glossary

Term	Definition
Solo Operator	Independent service provider working alone, managing 2-3 jobs per day without office staff
Estimate	A quoted price for services sent to a customer for approval before work begins; may include multiple options
Invoice	A bill sent to the customer after work is completed, requesting payment
Job / Work Order	A scheduled instance of work to be performed, including date, time, location, and description
Item	A reusable service or material entry with description, rate, and tax settings
Option (Estimate)	One of up to 3 pricing alternatives (shown as tabs) presented in an estimate
FSM	Field Service Management — software for managing field operations